# 1 Supplementary Methodological and Technical Appendix

**Manuscript Title:** Downside Risk Mitigation in Cryptocurrency Markets: A Dual-Stream Gated Hybrid LSTM Framework under Market Frictions
**Target Journal:** *Journal of Accounting and Investment* (JAI - SINTA 2 Accredited)
**Author(s):** [Redacted for Double-Blind Peer Review]
**Document Status:** Technical Supplement to Main Empirical Manuscript

---

## 1.1 Overview

This supplementary appendix provides comprehensive technical, algorithmic, diagnostic, and computational specifications to accompany the primary manuscript. It is designed to furnish quantitative reviewers, econometricians, and algorithmic researchers with the complete mathematical and operational blueprints necessary for rigorous independent verification and empirical replication.

The contents are organized into five dedicated technical sections: - **Appendix A:** Full Hyperparameter Search Grid, Layer Dimensions, Training Protocol, and Parameter Optimization Reconciliation - **Appendix B:** Formal Algorithmic Pseudocode of the Confidence Dead-Band Hysteresis Policy - **Appendix C:** Comprehensive Stationarity Diagnostic Logs and Unit Root Tests - **Appendix D:** Computational Hardware, Software Environment, and Runtime Execution Benchmarks - **Appendix E:** Stationary Block Bootstrap Resampling Methodology (Politis & Romano, 1994)

---

## 1.2 Appendix A: Hyperparameter Search Grid, Layer Dimensions, and Training Protocol

### 1.2.1 A.1. Network Architecture and Dimension Topology

The proposed **ETH-Sentinel Hybrid LSTM** comprises 9,681 trainable parameters structured across two isolated recurrent feature extraction streams, a Gated Cross-Modality Fusion Unit (GFU), and a calibrated binary classification head. Table A1 details the layer-by-layer dimension topology.

#### *1.2.1.1 Table A1. Detailed Layer Specifications and Tensor Geometries of the ETH-Sentinel Hybrid LSTM*

| Module Component | Layer Name | Layer Type | Output Dimension | Activation Function | Dropout Rate | Trainable Parameters |
|---|---|---|---|---|---|---|
| **Stream 1: Technical Price** | price_input | InputLayer | $(B, 18, 4)$ | None | None | 0 |
| | lstm_p1 | Stacked LSTM | $(B, 18, 24)$ | Tanh / Sigmoid | 0.20 (Recurrent: 0.10) | 2,784 |
| | lstm_p2 | Stacked LSTM | $(B, 16)$ | Tanh / Sigmoid | 0.20 | 2,624 |
| | ln_price | LayerNormalization | $(B, 16)$ | Linear | None | 32 |
| **Stream 2: Behavioral Sentiment** | sent_input | InputLayer | $(B, 18, 3)$ | None | None | 0 |

| Module Component | Layer Name | Layer Type | Output Dimension | Activation Function | Dropout Rate | Trainable Parameters |
|---|---|---|---|---|---|---|
| | lstm_s1 | Stacked LSTM | $(B, 18, 16)$ | Tanh / Sigmoid | 0.20 (Recurrent: 0.10) | 1,280 |
| | lstm_s2 | Stacked LSTM | $(B, 16)$ | Tanh / Sigmoid | 0.20 | 2,112 |
| | ln_sent | LayerNormalization | $(B, 16)$ | Linear | None | 32 |
| **GFU Multi-Modal Fusion** | concat_streams | Concatenate | $(B, 32)$ | None | None | 0 |
| | gfu_gate | Dense | $(B, 16)$ | Logistic Sigmoid | None | 528 |
| | gfu_fusion | Lambda (Convex Combination) | $(B, 16)$ | Linear | None | 0 |
| **Classification Head** | dense_head | Dense | $(B, 16)$ | ReLU | None | 272 |
| | head_dropout | Dropout | $(B, 16)$ | None | 0.20 | 0 |
| | output_prob | Dense | $(B, 1)$ | Logistic Sigmoid | None | 17 |
| **Total Model Capacity** | **Full Architecture** | — | — | — | — | **9,681 Parameters** |

*Note: B denotes dynamic batch size ($B = 64$ during training). Lookback window length is fixed at $W = 18$ bars (72 consecutive hours).*

---

## 1.2.2 A.2. Hyperparameter Grid Search and Training Regularization

Hyperparameter optimization was conducted via chronological validation on the validation set ($N = 636$ sequences) to identify configurations that minimize validation binary cross-entropy loss without overfitting. Table A2 reports the optimal hyperparameter specifications and exploration bounds.

### *1.2.2.1 Table A2. Hyperparameter Search Space and Optimal Configuration*

| Hyperparameter Parameter | Search Range Considered | Optimal Selected Value | Methodological Rationale |
|---|---|---|---|
| **Lookback Sequence Window ($W$)** | $\{6, 12, 18, 24, 30\}$ bars | **18 bars (72 hours)** | Captures the 3-day multi-session behavioral momentum cycle without introducing vanishing gradients. |
| **Optimizer Algorithm** | SGD, RMSprop, Adam, AdamW | **Adam** | Adaptive moment estimation ( $\beta_1 = 0.9, \beta_2 = 0.999,$ ). |
| **Initial Learning Rate ($\eta$)** | $\{10^{-4}, 5 \times 10^{-4}, 10^{-3}, 5 \times$ | **0.0010 ($10^{-3}$)** | Ensures rapid convergence while maintaining numerical gradient stability. |
| **Mini-Batch Size ( $B$)** | $\{32, 64, 128\}$ | **64** | Balances stochastic gradient variance and GPU |

| Hyperparameter Parameter | Search Range Considered | Optimal Selected Value | Methodological Rationale |
|---|---|---|---|
| | | | tensor throughput. |
| **Maximum Training Epochs** | $\{50, 100, 150\}$ | **100 Epochs** | Combined with strict early stopping to prevent over-parameterization. |
| **Early Stopping Patience** | $\{5, 10, 15\}$ epochs | **10 Epochs** | Monitors validation loss ( $L_{\text{val}}$); automatically restores global optimal weights. |
| **Feature Dropout Rate** | $\{0.10, 0.20, 0.30\}$ | **0.20** | Applied after LSTM output vectors and intermediate dense layers. |
| **Recurrent Dropout Rate** | $\{0.00, 0.10, 0.20\}$ | **0.10** | Regularizes hidden state transitions across consecutive time steps. |
| **Loss Function** | BCE, Focal Loss, Hinge | **Binary Cross-Entropy** | $L_{\text{BCE}} = -\frac{1}{N}\sum_{i=1}^{N}[\, y_i \ln$ . |
| **Feature Normalization** | MinMax, Standard, Robust | **Isolated RobustScaler** | Uses median and IQR fitted exclusively on training set to handle extreme |

| Hyperparameter Parameter | Search Range Considered | Optimal Selected Value | Methodological Rationale |
|---|---|---|---|
| | | | cryptocurrency outliers. |

### 1.2.3 A.3. Parameter Optimization Reconciliation: Validation Tuning vs. Out-of-Sample Test Evaluation

To guarantee that the reported empirical performance is completely free from test-set data snooping or post-hoc threshold selection bias (Aronson, 2006; White, 2000), the hysteresis confidence threshold ($\delta^*$) was optimized strictly on the chronological validation set ($N = 636$ bars) via the formal constrained optimization objective:

$$\delta^* = \arg\max_{\delta \in \Delta} \left| \text{Sharpe}_{\text{val}}(\delta) \right| \text{ subject to } \begin{cases} \text{MDD}_{\text{val}}(\delta) \leq \tau_{\text{MDD}} \\ K_{\text{val}}(\delta) \leq \tau_{\text{Turnover}} \end{cases}$$

where $\delta^* = 0.01$ was selected because it uniquely achieved maximum turnover compression with minimal Sharpe degradation (¿ 10 %) and lowest validation drawdown. Table A3 reports the full grid search exploration across $\delta \in [0.00, 0.10]$ alongside the locked out-of-sample evaluation on the test set ($N = 638$ bars).

*1.2.3.1 Table A3. Hysteresis Threshold Grid Optimization on Validation Set ($N = 636$ bars) and Out-of-Sample Test Evaluation ($N = 638$ bars)*

| Threshold ($\delta$) | Validation Set ($N = 636$ bars, May 2025 – Sep 2025) | | | | Out-of-Sample Test Set ($N = 638$ bars, Sep 2025 – Sep 2026) | | | | Operational Selection Status |
|---|---|---|---|---|---|---|---|---|---|
| | **Cumulative Return (%)** | **Sharpe Ratio** | **Maximum Drawdown (%)** | **Turnover (Trades)** | **Net Return (%)** | **Sharpe Ratio** | **Maximum Drawdown (%)** | **Turnover (Trades)** | |
| **Panel A: Proposed ETH-Sentinel Dual-Stream GFU Hybrid Model** | | | | | | | | | |
| $\delta = 0.00$ | +23.94% | 2.23 | −13.28% | 70 | +1.77% | 0.35 | −18.82% | 40 | Unco |

| | Validation Set ($N = 636$ bars, May 2025 – Sep 2025) | | | Out-of-Sample Test Set ($N = 638$ bars, Sep 2025 – Sep 2026) | | | | Operational Selection Status |
|---|---|---|---|---|---|---|---|---|
| Threshold ($\delta$) | | | | | | | | |
| | | | | | | | | nstrained Sharpe Peak (High Turnover Drag) |
| $\delta^* = 0.01$ | +21.70% | 2.03 | −12.17% | 23 | +15.99% | 1.42 | −18.19% | 16 | **Optimal Constrained Policy (MinMDD, -67** |

| Threshold ($\delta$) | Validation Set ($N = 636$ bars, May 2025 – Sep 2025) | | | | Out-of-Sample Test Set ($N = 638$ bars, Sep 2025 – Sep 2026) | | | | Operational Selection Status |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | **% Trades)** |
| $\delta = 0.02$ | −0.86% | 0.08 | −19.80% | 9 | +2.41% | 0.43 | −18.33% | 8 | Sub-optimal Filter |
| $\delta = 0.03$ | −4.93% | −0.38 | −22.63% | 3 | +14.94% | 1.38 | −16.95% | 3 | Over-dampened Filter |
| $\delta = 0.04$ | +3.39% | 0.48 | −19.03% | 1 | +33.33% | 2.33 | −16.95% | 3 | Post-Hoc Sensitivity Landscape (Ablation |

| Threshold ($\delta$) | Validation Set ($N = 636$ bars, May 2025 – Sep 2025) | | | | Out-of-Sample Test Set ($N = 638$ bars, Sep 2025 – Sep 2026) | | | | Operational Selection Status |
|---|---|---|---|---|---|---|---|---|---|
| $\delta = 0.05$ | +2.77% | 0.43 | −19.03% | 1 | +33.33% | 2.33 | −16.95% | 3 | Boundary Conviction Limit Only) |
| $\delta \geq 0.06$ | 0.00% | 0.00 | 0.00% | 0 | 0.00% | 0.00 | 0.00% | 0 | Complete Cash (Abstention) |
| **Panel B: Ablation Benchmark: Techni** | | | | | | | | | |

| Threshold ($\delta$) | Validation Set ($N = 636$ bars, May 2025 – Sep 2025) | | | | Out-of-Sample Test Set ($N = 638$ bars, Sep 2025 – Sep 2026) | | | | Operational Selection Status |
|---|---|---|---|---|---|---|---|---|---|
| **cal 2-Layer LSTM Baseline** | | | | | | | | | |
| $\delta = 0.00$ | −6.66% | −0.27 | −16.90% | 33 | +12.33% | 1.11 | −24.42% | 39 | Raw Technical Model Baseline |
| $\delta^* = 0.01$ | +9.51% | 0.86 | −16.49% | 13 | +15.18% | 1.29 | −17.24% | 13 | **Optimal Validation Choice for Technical** |

| Threshold ($\delta$) | Validation Set ($N = 636$ bars, May 2025 – Sep 2025) | | | | Out-of-Sample Test Set ($N = 638$ bars, Sep 2025 – Sep 2026) | | | | Operational Selection Status |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | **LSTM** |
| $\delta = 0.02$ | +6.20% | 0.67 | −17.94% | 5 | +2.78% | 0.44 | −18.34% | 5 | Sub-optimal Filter |
| $\delta \geq 0.03$ | +3.87% to 0.00% | $\leq 0.51$ | −19.03% | $1 \rightarrow 0$ | 0.00% | 0.00 | 0.00% | 0 | Technical Signal Freezes into Cash |

*Note: In accordance with econometric best practices against data snooping, $\delta = 0.00$ represents the unconstrained validation Sharpe peak (2.23), but $\delta^*_{\text{val}} = 0.01$ was selected as the optimal constrained operating policy on the validation partition prior to any test set evaluation. On the validation partition, $\delta^* = 0.01$ achieved the lowest Maximum Drawdown ( −12.17%) and reduced turnover by 67.1% (from 70 to 23 trades) with only a 9.0% discount in the Sharpe ratio (2.23 → 2.03). When transferred to the unseen out-of-sample test set, this parameter achieved +15.99% net compounded return (Sharpe 1.42, MDD −18.19%, 16*

*trades), outperforming the technical baseline (+15.18%, Sharpe 1.29). The higher return at* $\delta = 0.04$ *on the test set is strictly documented as a post-hoc sensitivity exploration and is not claimed as an ex-ante operational result.*

---

## 1.3 Appendix B: Formal Algorithmic Specification of Confidence Dead-Band Hysteresis Backtesting

The complete backtesting procedure, incorporating the institutional transaction friction fee ( $\phi = 10\,\text{bps} = 0.0010$) and the Confidence Dead-Band Hysteresis Policy ($\delta \in [\,0.00\,, 0.10\,]$), is formally documented in **Algorithm 1**.

==========

Algorithm 1: Out-of-Sample Confidence Dead-Band Hysteresis Backtesting Engine

==========

Input:

- Trained Dual-Stream Gated Hybrid Model: M_theta
- Out-of-Sample Price Sequences: X_price in R^(T x W x 4)
- Out-of-Sample Sentiment Sequences: X_sent in R^(T x W x 3)
- Next-Bar Actual Arithmetic Asset Returns: R_asset in R^T, where R_{t+1}^asset = exp(r_{t+1}) - 1
- Transaction Friction Coefficient: phi = 0.0010 (10 basis points per position transition)
- Policy Confidence Half-Width Margin: delta in [0.00, 0.10]
- Cryptocurrency Annualization Factor: AF = 2,190 (365 days x 6 four-hour bars/day)

Output:

- Cumulative Net Strategy Return: Net_Return (%)
- Annualized Sharpe Ratio: Sharpe
- Maximum Portfolio Drawdown: MDD (%)
- Total Turnover Executions: K
- Total Transaction Fee Friction Drag: Friction (%)
- Portfolio Equity Curve: V = {V_0, V_1, ..., V_T}

Procedure:

1: Initialize portfolio equity: V_0 <- 1.0

```
Initialize prior position state: S_{−1} <- 0.0 (Neutral / 100% Cash)
Initialize turnover counter: K <- 0
Initialize cumulative friction cost: C_total <- 0.0
Initialize empty return array: R_net <- zeros(T)

For each bar t = 0 to T - 1 do
    // Generate posterior upward probability from trained model
    P_hat_t <- M_theta(X_price[t], X_sent[t]) in (0, 1)

    // Apply Confidence Dead-Band Hysteresis Rule
    If P_hat_t > 0.50 + delta then
        S_t <- 1.0                      // Long Position
    Else if P_hat_t < 0.50 - delta then
        S_t <- 0.0                      // Neutral / Cash Position
    Else
        S_t <- S_{t-1}                  // Inertia: Retain previous state
    End If

    // Calculate transaction friction cost incurred at bar t
    turnover_indicator <- |S_t - S_{t-1}|
    c_t <- turnover_indicator * phi
    If turnover_indicator > 0 then
        K <- K + 1
        C_total <- C_total + c_t
    End If

    // Compute net realized strategy return over forward holding interval (t, t+1]
    R_net[t] <- S_t * R_asset[t] - c_t

    // Iterative wealth accumulation
    V_{t+1} <- V_t * (1.0 + R_net[t])
End For

// Performance Metrics Computation
Net_Return <- (V_T - 1.0) * 100.0
mean_ret <- mean(R_net)
```

```
std_ret <- std(R_net, ddof=1)
If std_ret > 0 then
    Sharpe <- (mean_ret / std_ret) * sqrt(AF)
Else
    Sharpe <- 0.0
End If

// Maximum Drawdown Computation
running_max_0 <- V_0
max_dd <- 0.0
For tau = 1 to T do
    running_max_tau <- max(running_max_{tau-1}, V_tau)
    dd_tau <- (V_tau - running_max_tau) / running_max_tau
    If dd_tau < max_dd then
        max_dd <- dd_tau
    End If
End For
MDD <- max_dd * 100.0
Friction <- C_total * 100.0

Return {Net_Return, Sharpe, MDD, K, Friction, V}
```

---

## 1.4 Appendix C: Comprehensive Stationarity Diagnostic Logs and Unit Root Tests

To establish empirical rigor and guarantee that non-stationary unit roots do not induce spurious econometric correlations in the recurrent networks, formal **Augmented Dickey-Fuller (ADF)** tests were conducted across all raw and engineered variables.

### 1.4.1 C.1. Econometric Test Formulation

The ADF regression with a drift constant and zero deterministic trend is formulated as:

$$\Delta y_t = \alpha + \beta t + \gamma y_{t-1} + \sum_{i=1}^{p} \delta_i \Delta y_{t-i} + \varepsilon_t$$

where the null hypothesis testing unit-root non-stationarity is:

$$H_0 : \gamma = 0 \,(\text{Non-Stationary, Unit Root Exists})$$

versus the alternative hypothesis of stationarity:

$$H_1 : \gamma < 0 \,(\text{Stationary, Integrated of Order Zero: } I(0))$$

The optimal lag length $p$ is selected automatically by minimizing the Akaike Information Criterion (AIC).

## 1.4.2 C.2. Empirical ADF Diagnostic Results

Table C1 reports the numerical test statistics, $p$-values, critical values, and integration orders across all candidate variables over the full 4,370-bar historical sample (September 18, 2024 to September 17, 2026).

### *1.4.2.1 Table C1. Augmented Dickey-Fuller Unit Root Test Diagnostics ($N = 4,370$ 4-Hour Bars)*

| Variable Code | Variable Description | ADF Test Statistic | $p$-Value | 1% Critical Value | 5% Critical Value | 10% Critical Value | Integration Order | Econometric Decision |
|---|---|---|---|---|---|---|---|---|
| $P_t$ | Raw Ethereum Closing Price | $-1.4815$ | 0.5426 | $-3.4318$ | $-2.8622$ | $-2.5672$ | $I(1)$ | Fail to Reject $H_0$ (Non-Stationary) |
| $r_t$ | 4-Hour Log-Return ( | $-64.7126$ | ¿ $1.0 \times 10^{-}$ | $-3.4318$ | $-2.8622$ | $-2.5672$ | $I(0)$ | **Reject $H_0$ at 1% Level (Station** |

| Variable Code | Variable Description | ADF Test Statistic | $p$-Value | 1% Critical Value | 5% Critical Value | 10% Critical Value | Integration Order | Econometric Decision |
|---|---|---|---|---|---|---|---|---|
| | $\ln(P_t/I$ ) | | | | | | | **ary)** |
| $\sigma_{GK,t}^{\text{ann}}$ | Garman-Klass Volatility (20-bar) | $-7.8924$ | $4.38\times10^{-}$ | $-3.4318$ | $-2.8622$ | $-2.5672$ | $I(0)$ | **Reject $H_0$ at 1% Level (Stationary)** |
| $\sigma_{CC,t}$ | Close-to-Close Rolling Volatility | $-8.1402$ | $1.01\times10^{-}$ | $-3.4318$ | $-2.8622$ | $-2.5672$ | $I(0)$ | **Reject $H_0$ at 1% Level (Stationary)** |
| $\text{RSI}_{14}$ | 14-Period Relative Strength Index | $-12.4518$ | $3.55\times10^{-}$ | $-3.4318$ | $-2.8622$ | $-2.5672$ | $I(0)$ | **Reject $H_0$ at 1% Level (Stationary)** |
| $S_t$ | Normalized CFGI Score ( $F_d/100$ | $-5.1240$ | $1.25\times10^{-}$ | $-3.4318$ | $-2.8622$ | $-2.5672$ | $I(0)$ | **Reject $H_0$ at 1% Level (Station** |

| Variable Code | Variable Description | ADF Test Statistic | $p$-Value | 1% Critical Value | 5% Critical Value | 10% Critical Value | Integration Order | Econometric Decision |
|---|---|---|---|---|---|---|---|---|
| | ) | | | | | | | **ary)** |
| $\text{EMA}_{S,t}$ | 24-Hour Exponential Decay Score | $-4.9811$ | $2.42 \times 10^{-}$ | $-3.4318$ | $-2.8622$ | $-2.5672$ | $I(0)$ | **Reject $H_0$ at 1% Level (Stationary)** |
| $\Delta S_t$ | Daily Sentiment Momentum ( $S_t - S_{t-1}$ ) | $-24.8912$ | ¿ $1.0 \times 10^{-}$ | $-3.4318$ | $-2.8622$ | $-2.5672$ | $I(0)$ | **Reject $H_0$ at 1% Level (Stationary)** |

*Diagnostic Conclusion:* While the raw price level $P_t$ is non-stationary ($p = 0.5426$), all eight engineered technical and sentiment features decisively reject the null hypothesis of a unit root at the 1% significance level ($p < 0.0001$). This verifies that the input tensors feeding both recurrent streams of the ETH-Sentinel Hybrid LSTM are strictly stationary ($I(0)$), eliminating spurious temporal regressions.

---

## 1.5 Appendix D: Computational Hardware, Software Stack, and Runtime Profiling

To ensure transparency regarding computational resource requirements and demonstrate institutional deployability, Table D1 outlines the complete hardware environment, software specifications, and empirical runtime profiling.

*1.5.0.1 Table D1. Computational Environment and Execution Runtime Profiling*

| Category | Component Specification | Detailed Measurement / Configuration |
|---|---|---|
| **Hardware Infrastructure** | Processor Architecture | Apple M-Series Silicon (8-Core High-Performance / Efficiency CPU) |
| | Graphics Accelerator | 10-Core Integrated GPU (Metal Performance Shaders acceleration) |
| | System Memory (RAM) | 16 GB Unified LPDDR5 Memory |
| | Storage Drive | PCIe Gen 4 Solid State Drive (Read: 3,200 MB/s, Write: 2,900 MB/s) |
| **Software Platform** | Operating System | macOS Darwin 24.6.0 (64-bit Unix Kernel) |
| | Programming Language | Python 3.9.6 (C-Python Distribution) |
| | Core Machine Learning Framework | TensorFlow 2.20.0 / Keras 3.10.0 |
| | Numerical & Data Libraries | NumPy 2.0.2, Pandas 2.3.3, Scikit-Learn 1.6.1 |
| | Econometric & Statistics | Statsmodels 0.14.6 |
| | Reproducibility Environment | Deterministic seeds frozen (SEED = 42, PYTHONHASHSEED = 42) |
| **Empirical Runtime Profiling** | Pipeline Step 1: CFGI Ingestion (01) | 1.45 seconds (3,147 daily records fetched via REST API) |
| | Pipeline Step 2: Feature Engineering (02) | 2.82 seconds (4,370 bars processed, ADF tested, zero |

| Category | Component Specification | Detailed Measurement / Configuration |
|---|---|---|
| | | NaN) |
| | Pipeline Step 3: Sequence Construction (03) | 1.76 seconds (Tensor construction: Train 3,032, Val 636, Test 638) |
| | Pipeline Step 4: Baseline Training (04) | 18.34 seconds (Linear Ridge, Random Forest, Technical LSTM) |
| | Pipeline Step 5: Hybrid Sentinel Training (05) | 22.48 seconds (Average: 1.25 s/epoch; converged at Epoch 18) |
| | Pipeline Step 6: Sensitivity Analysis (06) | 1.12 seconds (11 hysteresis dead-band policies evaluated) |
| | **Total End-to-End Execution Time** | **47.97 seconds (Under 1 minute total compute)** |
| **System Memory Footprint** | Peak Resident Memory (RSS) | **418 MB RAM** |
| | Disk Footprint (Code + Data + Models) | **14.2 MB Total** |

*Operational Takeaway:* The entire research pipeline requires less than 48 seconds of execution time and consumes under 420 MB of RAM, proving that the ETH-Sentinel Hybrid architecture can be effortlessly deployed in real-time institutional production environments, continuous retraining pipelines, or lightweight edge nodes without necessitating costly multi-GPU enterprise infrastructure.

## 1.6 Appendix E: Stationary Block Bootstrap Resampling Methodology (Politis & Romano, 1994)

To rigorously evaluate the statistical significance of empirical performance differentials between the ETH-Sentinel GFU Hybrid model and the standalone Technical LSTM benchmark, we implement the **Stationary Block Bootstrap** developed by Politis and Romano (1994). This non-parametric resampling framework is uniquely suited for high-frequency financial time series exhibiting temporal dependence, conditional heteroskedasticity (volatility clustering), and non-Gaussian return distributions.

### 1.6.1 E.1. Theoretical Formulation and Geometric Block Sampling

Standard independent and identically distributed (i.i.d.) bootstrap resampling (Efron, 1979) destroys the serial correlation and volatility dynamics inherent in cryptocurrency returns. While moving-block bootstrap methods (Künsch, 1989) preserve localized dependencies, they introduce artificial non-stationarity because blocks have fixed boundaries.

The stationary bootstrap of Politis and Romano (1994) resolves this limitation by wrapping the time series circularly and sampling pseudo-series through blocks of random, geometrically distributed length. Let $\{R_t\}_{t=0}^{T-1}$ represent the realized strategy return series on the out-of-sample test partition ($T = 638$ bars). The circular time index is defined such that $R_t = R_{t \bmod T}$ for all $t \geq T$.

The algorithm generates a bootstrap pseudo-series $\{R_t^*\}_{t=0}^{T-1}$ as follows: 1. **Initial Index Selection:** The first element index $I_0$ is drawn uniformly at random from $\{0, 1, \ldots, T-1\}$, and we set $R_0^* = R_{I_0}$. 2. **Sequential Dynamic Transition:** For each subsequent time step $t = 1, 2, \ldots, T-1$, a Bernoulli trial with success probability $p \in (0, 1)$ governs the transition:

$$I_t = \begin{cases} (I_{t-1} + 1) \bmod T & \text{with probability } 1 - p \\ U_t \sim \text{Uniform}(\{0, 1, \ldots, T-1\}) & \text{with probability } p \end{cases}$$

where $U_t$ is an independent discrete uniform random variable. Consequently, the length of each contiguous data block $L_k$ follows a geometric distribution:

$$P(L_k = m) = (1-p)^{m-1} p, m \in \{1, 2, 3, \ldots\}$$

with expected mean block length $E[L] = b = \frac{1}{p}$.

### 1.6.2 E.2. Block Length Calibration and Test Horizon Alignment

In our empirical implementation, the mean block length is calibrated to:

$$b = 18 \text{ bars (equivalent to } p = \frac{1}{18} \approx 0.0556)$$

This parameter is economically and mathematically grounded in two structural considerations: 1. **Model Lookback Alignment:** The temporal sequence length feeding the LSTM recurrent cells is exactly $W = 18$ bars (72 consecutive hours / 3 full diurnal trading cycles). Choosing $b = 18$ ensures that the resampled synthetic paths preserve the multi-day momentum memory and autoregressive dependency structures captured by the neural network. 2. **Empirical Autocorrelation Decay:** Analysis of the autocorrelation function (ACF) and partial autocorrelation function (PACF) of ETH squared log-returns indicates that short-term volatility clustering decays substantially after 16 to 20 four-hour intervals.

### 1.6.3 E.3. Bootstrap Test Statistics and Confidence Interval Estimation

We generate $B = 1,000$ independent stationary bootstrap pseudo-series pairs:

$$\left\{ \left( R^{¿(b)}_{\text{Hybrid},t}, R^{¿(b)}_{\text{Technical},t} \right) \right\}_{t=0}^{T-1}, b \in \{1, 2, \ldots, B\}$$

Using identical index sequences for both models within each bootstrap resample $b$ guarantees that the cross-sectional covariance and common market shock structures are strictly preserved.

For each bootstrap replication $b$, we compute the annualized Sharpe ratio ($\text{SR}^{¿(b)}$) and Maximum Drawdown ($\text{MDD}^{¿(b)}$) for both architectures, yielding the empirical differential distributions:

$$\Delta \widehat{\text{SR}}^{¿(b)} = \text{SR}^{¿(b)}_{\text{Hybrid}} - \text{SR}^{¿(b)}_{\text{Technical}}$$

$$\Delta \widehat{\text{MDD}}^{¿(b)} = \text{MDD}^{¿(b)}_{\text{Hybrid}} - \text{MDD}^{¿(b)}_{\text{Technical}}$$

The two-sided empirical 95% confidence interval for any parameter difference $\Delta\theta \in \{\Delta\text{SR}, \Delta\text{MDD}\}$ is constructed via the empirical percentile method:

$$\mathrm{CI}_{0.95} = \left[ \Delta\hat{\theta}^{*}_{(0.025)}, \Delta\hat{\theta}^{*}_{(0.975)} \right]$$

where $\Delta\hat{\theta}^{*}_{(\alpha)}$ denotes the $\alpha$-quantile of the $B$ bootstrap estimates.

### 1.6.4 E.4. Hypothesis Testing under the Ledoit & Wolf (2008) Framework

To test the formal null hypothesis of zero difference in risk-adjusted performance:

$$H_0 : \mathrm{SR}_{\text{Hybrid}} - \mathrm{SR}_{\text{Technical}} = 0 \text{ versus } H_1 : \mathrm{SR}_{\text{Hybrid}} \neq \mathrm{SR}_{\text{Technical}}$$

we implement the centered studentized bootstrap testing framework of Ledoit and Wolf (2008). Under the null hypothesis $H_0$, the bootstrap distribution is centered around zero by subtracting the observed point estimate on the original sample:

$$z^{¿(b)} = \Delta\widehat{\mathrm{SR}}^{¿(b)} - \Delta\widehat{\mathrm{SR}}_{\text{obs}}$$

The empirical two-sided bootstrap $p$-value is given by:

$$p = \frac{1}{B}\sum_{b=1}^{B} I\left( \left| z^{¿(b)} \right| \geq \left| \Delta\widehat{\mathrm{SR}}_{\text{obs}} \right| \right)$$

where $I(\cdot)$ is the indicator function.

#### *1.6.4.1 Table E1. Stationary Block Bootstrap Empirical Test Diagnostics ($B = 1,000$ Replications, $b = 18$ Bars)*

| Evaluation Policy Setting | Performance Metric Comparison | Observed Sample Difference ($\Delta_{\text{obs}}$) | Bootstrap Mean Difference | Bootstrap Standard Error | Two-Sided 95% Confidence Interval | Ledoit-Wolf Bootstrap $p$-Value | Econometric Interpretation |
|---|---|---|---|---|---|---|---|
| **Raw Baseline ($\delta = 0.00$)** | Annualized Sharpe Ratio | $-0.75$ | $-0.78$ | 0.91 | $[-2.61, +0.$ | $p = 0.4310$ | Fail to Reject $H_0$; high turnover noise in both models |

| Evaluation Policy Setting | Performance Metric Comparison | Observed Sample Difference ($\Delta_{obs}$) | Bootstrap Mean Difference | Bootstrap Standard Error | Two-Sided 95% Confidence Interval | Ledoit-Wolf Bootstrap $p$-Value | Econometric Interpretation |
|---|---|---|---|---|---|---|---|
| | Maximum Drawdown (MDD) | +5.60% pts | +5.61% pts | 4.49% pts | [−9.25%,+ | — | Tail-risk compressed from −24.42% to −18.82% (22.9% reduction) |
| **Hysteresis Locked ($\delta^{*}=0.01$)** | Annualized Sharpe Ratio | +0.13 | +0.09 | 1.21 | [−2.32,+2. | $p=0.9170$ | Statistically comparable point returns with lower turnover |
| | Maximum Drawdown (MDD) | −0.95% pts | −0.98% pts | 5.76% pts | [−10.28%, | — | Empirical probability of drawdown mitigati |

| Evaluation Policy Setting | Performance Metric Comparison | Observed Sample Difference ($\Delta_{obs}$) | Bootstrap Mean Difference | Bootstrap Standard Error | Two-Sided 95% Confidence Interval | Ledoit-Wolf Bootstrap $p$-Value | Econometric Interpretation |
|---|---|---|---|---|---|---|---|
| | | | | | | | on $P(\Delta \text{MD}$ |

The empirical results in Table E1 establish that while the Sharpe ratio point estimate of the Hybrid model at $\delta^* = 0.01$ is higher than the Technical baseline (1.42 vs. 1.29, $\Delta \text{Sharpe} = +0.13$), the difference is not statistically distinguishable from zero at conventional asymptotic significance levels ($p = 0.9170$). This wide estimation interval ($[-2.32, +2.45]$) reflects the finite effective sample size: across $T = 638$ test intervals, an expected geometric block length of $b = 18$ bars yields approximately $N_{\text{eff}} = 638/18 \approx 35.4 \approx 35$ independent temporal blocks. Furthermore, at $\delta^* = 0.01$, the Maximum Drawdown of the Hybrid model ($-18.19\%$) is slightly higher by 0.95 percentage points compared to the Technical model ($-17.24\%$), demonstrating that once the hysteresis filter is engaged, decision inertia itself acts as the primary structural cushion against drawdown for both architectures. Under active hysteresis, the comparative advantage of the behavioral sentiment stream transitions from pure drawdown compression (which operated dominantly at $\delta = 0.00$ with 5.60 percentage points protection) to net alpha expansion (+15.99% vs. +15.18%) and risk-adjusted efficiency.

### 1.6.5 E.5. Algorithmic Implementation and Computational Reproducibility

The bootstrap engine is natively integrated in Code/07_remedial_audit_experiments.py. Across all $1,000$ iterations, reproducibility is guaranteed by fixing the seed (SEED = 42). Total execution runtime for the complete 1,000-replication stationary bootstrap on an Apple M-series CPU is 14.62 seconds, requiring negligible additional memory overhead (¿15 MB).

---

## 1.7 References (Supplementary Appendix)

Aronson, D. R. (2006). *Evidence-based technical analysis: Applying the scientific method and statistical inference to trading signals*. John Wiley & Sons.

Efron, B. (1979). Bootstrap methods: Another look at the jackknife. *The Annals of Statistics*, 7(1), 1–26. https://doi.org/10.1214/aos/1176344552

Hendrickx, K., Perini, L., Van der Plas, D., Meert, W., & Davis, J. (2024). Machine learning with a reject option: A survey. *Machine Learning*, 113(5), 3073–3110. https://doi.org/10.1007/s10994-024-06534-x

Künsch, H. R. (1989). The jackknife and the bootstrap for general stationary observations. *The Annals of Statistics*, 17(3), 1217–1241. https://doi.org/10.1214/aos/1176347265

Ledoit, O., & Wolf, M. (2008). Robust performance hypothesis testing with the Sharpe ratio. *Journal of Empirical Finance*, 15(5), 850–859. https://doi.org/10.1016/j.jempfin.2008.03.002

Lo, A. W. (2002). The statistics of Sharpe ratios. *Financial Analysts Journal*, 58(4), 36–52. https://doi.org/10.2469/faj.v58.n4.2453

Politis, D. N., & Romano, J. P. (1994). The stationary bootstrap. *Journal of the American Statistical Association*, 89(428), 1303–1313. https://doi.org/10.1080/01621459.1994.10476870

White, H. (2000). A reality check for data snooping. *Econometrica*, 68(5), 1097–1126. https://doi.org/10.1111/1468-0262.00152